PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

DAR ES SALAAM REGION

FORM FOUR - MOCK EXAMINATION - 2025

CHEMISTRY 2B

(ACTUAL PRACTICAL)

CODE:032/2B

TIME:2:30 Hours — Tuesday 29th July, 2025 A.M.

INSTRUCTIONS

1. This paper consists of **two (2)** questions.
2. Answer **All** questions
3. Each question carries **twenty-five (25)** marks
4. Cellular phones and any unauthorized materials are **not** allowed in the examination room.
5. Qualitative analysis guide sheet authorized by a supervisor may be used
6. The following constants may be used
   Atomic masses; H = 1, O = 16, Na = 23, Cl = 35.5

1. A sample solution of 0.1m HCl (T T) was titrated with a base solution made by dissolving 1.39g of impure anhydrous Na_2CO_3 (YY) to make 250cm³ solution. Using POP indicator, titrate solution T T against solution YY to determine the amount of acid required to complete the reaction and lastly answer the questions that follows.

- POP indicator my be used.

PROCEDURE

(i) Titrate solution T T against 20cm³ or 25cm³ of YY until the colour change. Record the burette readings.

(ii) Repeat the procedure (i) to obtain three accurate readings and record your results in a tabular form.

QUESTIONS

(a) (i) Why did the colour of the solution changed?

(ii) Determine the average titre volume

(iii) ________cm³ of solution YY required ________cm³ of solution T T for complete reaction.

(iv) Assume that sulphuric acid of the same molarity was used in the place of hydrochloric acid, would it be a difference in the titrate volume used? Give reason

(b) (i) Name the apparatus you used for measuring volume of T T.

(ii) Why the apparatus in b (i) is the best recommended for its functions?

(c) Write a balanced chemical equation of the reaction between the solution TT and YY.

(d) Calculate the following and write your answer in two decimal place

(i) Molarity of YY

(ii) Percentage purity of YY

(e) State two applications of volumetric analysis

2. Sample NN consists of one cation and one anion. Carry out systematic qualitative analysis procedures to identify the cation an anion in sample NN. Record carefully your experiment, observations and inferences as indicated in the experimental table.

EXPERIMENTAL TABLE

S/N	EXPERIMENTS	OBSERVATIONS	INFERENCES

Questions

(a) What are the cation and anion in the sample

(b) Write the formula of the sample

(c) Mention two properties of the cation identified

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MARKING SCHEME

TABLE OF RESULTS

Titration	Pilot	1	2	3
Final readings (cm^3)	25.40	24.50	25.00	49.50
Initial readings (cm^3)	0.00	0.00	0.00	25.00
Volume used (cm^3)	25.40	24.50	25.00	24.50

5 Marks

(I) The colour changed due to the end point at which the reaction was completed.

(II) Average titre volume $= \frac{T_1 + T_2 + T_3}{3}$ ½ Marks

$= \frac{(24.50 + 25.00 + 24.50)}{3} cm^3$

$= \frac{74 cm^3}{3}$ ½ marks

∴ Average titre volume $= 24.67 cm^3$

1 mark

02 Marks

a (iii) 25 cm^3 of solution YY required 24.67 cm^3 02 Marks
of solution TT for complete reaction

(iv) There will be difference in titre volume used. This is because the number of moles of acid (na) will decrease 01 Marks

b (i) Burette. 01 Marks

(ii) - It measure accurate volume of liquids
- Its scale start from the top of the apparatus to measure the volume of liquid which is being used. 01 Marks

c Balanced equation

$Na_2CO_{3(aq)} + 2HCl_{(aq)} \longrightarrow 2NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}$

02 Marks

d (i) Molarity

Data given

Mass of YY = 1.39 g 0½ Marks

Volume of YY = 250 cm^3

$$\text{Molarity} = \frac{\text{Concentration of YY in g/dm}^3}{\text{Molar mass of YY (g/mol)}}$$ 0½ Marks

$$\text{Concentration of YY in g/dm}^3 = \frac{\text{Mass (g)}}{\text{Volume (dm}^3)}$$

volume of y in dm^3

$1dm^3 = 1000cm^3$

$x dm^3 = 250cm^3$

$$x dm^3 = \frac{1dm^3 \times 250cm^3}{1000cm^3} = 0.25dm^3$$

0 ½ Marks

$$\therefore \text{Concentration of } YY = \frac{1.39g}{0.25dm^3}$$

$$= 5.56g/dm^3.$$

01 Marks

$$\text{Molar mass of } YY \ (Na_2CO_3) = (23 \times 2) + 12 + (16 \times 3)$$

$$= 46 + 12 + 48$$

$$= 106 gmol^{-1}$$

01 Marks

$$\text{Molarity of } YY = \frac{5.56g/dm^3}{106 gmol^{-1}}$$

$$= 0.05M.$$

$\therefore$ Molarity of YY $= 0.05M$ 01 Marks

(II) Percentage purity of YY

$$\text{Percentage purity} = \frac{\text{Concentration of pure } YY}{\text{Concentration of Impure } YY} \times 100\%$$

0 ½ Marks

Concentration of impure YY = 5.56 g/dm³

Concentration of pure YY = Molarity of pure YY x Molar mass of YY

Molarity of pure YY

given

- Molarity of acid (Ma) = 0.1M
- Molarity of base (Mb) = ?
- Volume of acid (Va) = 24.67 cm³ 0½ Marks
- Volume of base (Vb) 25cm³
- number of moles of acid (na) = 2
- number of moles of base (nb) = 1

From

$$\frac{MaVa}{MbVb} = \frac{na}{nb}$$ 0½ Marks

$$Mb = \frac{MaVa\,nb}{Vb\,na}$$

$$= \frac{0.1M \times 24.67\,cm^3 \times 1}{25cm^3 \times 2}$$

$$= 0.0493M$$ 01 Marks

Concentration of pure YY = 0.0493M x 106 g mol⁻¹

$= 5.2258 \simeq 5.23 g/dm^3$

$$\text{Percentage purity of YY} = \frac{5.23 g/dm^3}{5.56 g/dm^3} \times 100\%$$

0½ Marks

$$= 94.06\%$$

∴ Percentage purity of YY = 94.06%

01 Marks

2. EXPERMENTAL TABLE

S/N	Experment	Observations	Inferences
(a)	Appearance of sample	Blue crystals absor water from the atmosphere 2 Marks	Cu^{2+} may be present NO_3^-, SO_4^{2-}, Cl^- may be present
(b)	A glass rod or nicrome wire was dipped into concentrated HCl then heated in a non-luminous flame.	Bluish-green flame	Cu^{2+} may be present 2 Marks
(c)	A small amount of a solid sample into a dry test tube was heated gently then strongly until no further change	- Colourless gas with pungent smell evolved which turns moist blue litmus paper red or filter paper dipped in	SO_4^{2-} may be present 2 Marks

2.	Experment	Observations	Inferences
(c)		acidified potassium dichromate solution changed from yellow to green.	
		Black residue	Cu^{2+} may be present
(d)	A small amount of solid sample was transpered into a test tube, pollowed by 3 drops of dilute HCl	No gas evolved	SO_4^{2-}, NO_3^-, Cl^{-1} may be present 2 Marks
(e)	A small amount of a sample was transferred into a clean dry test tube, small amount of concentrated H_2SO_4 was added	Blue crystals which turn white without heating 2 Marks	SO_4^{2-} of hydrated Cu^{2+} may be present
f	A small amount of solid sample was transpered into the test tube then added with enough cold distilled water. The solution was divided into three portions	Soluble forming blue solution	Cu^{2+} may be present. 2 Marks

S/N	Experment	Observations	Inference
	(i) To one portion sodium hydroxide (NaOH) was added drow-wise until in excess	Blue precipitate was formed insoluble in excess.	Cu^{2+} may be present 2Marks
	(ii) To the second portion, ammonia solution was added drow-wise until in excess	Pale blue precipitate was formed soluble in excess ammonia forming a deep blue solution	Cu^{2+} Confirmed 2Marks
	(iii) To the third portion, barium chloride was added, followed by dilute HCl or barium nitrate followed by dilute HNO_3	White precipitate insoluble in dilute HCl or dilute HNO_3 was formed	SO_4^{2-} Confirmed 2Marks

Conclusions

(a) The cation in sample NN is Cu^{2+} and anion is SO_4^{2-} 1@ total 2 Marks

(b) Molecular formular is $CuSO_4$ 2Marks

(c) Properties of the cation (Cu^{2+}):-

- It is capable of easily accepting and donating electrons and forming a redox-reactions.